AI–IoT Based Early Risk Detection System for NAFLD, Diabetes, and Hypertension
Project Proposal

Team Name	Code-v
Category	University Category
Theme	Empowering SDGs through AIoT and Industry 5.0.

Problem Definition

Introduction

Introduction Our team identified this issue through our day-to-day observations and personal experiences. In Sri Lanka, we constantly see people we know—family members, friends, and neighbors—getting hit with serious health issues like diabetes or high blood pressure completely out of nowhere. The scary part is that these diagnoses usually happen way too late, once the permanent damage is already done. People are just going about their busy lives with no early warning system to tell them something is wrong before it becomes a medical emergency.

Identified Problem The specific issue we want to fix is how health data is stuck in separate pieces, or "data silos." Right now, if you care about your health, you might check your weight on a scale or look at your steps on a watch, but nothing connects them. There is a huge gap because nobody is combining things like your BMI, how well you sleep, and your heart rate to see the bigger picture. We want to bridge this gap to estimate the combined risk of fatty liver (NAFLD), type 2 diabetes, and hypertension. These three are the main parts of metabolic syndrome, but because they aren't tracked together, the early warning signs are missed.

Problem Analysis

Significance and Impact Metabolic syndrome is becoming a huge crisis in Sri Lanka. It’s estimated that nearly a quarter of adults here are affected, and most don't even know it. This isn't just a health issue; it impacts every part of our lives because it hits the productivity of people at work and at home. When people don’t realize their risk levels are going up, they end up with long-term illnesses that could have been prevented. This puts a massive, unnecessary burden on our families and the whole national healthcare system.

Industry 5.0 and AIoT Our solution is about moving past just basic automation. We want to follow Industry 5.0 by keeping things human-centric. Instead of a machine that just spits out a random number, our system acts like a partner to the person using it. It helps busy people, whether they are working or studying, to actually understand what their own body data means. This partnership between the user and the AI makes sure that technology is actually improving well-being instead of just being another cold clinical test.

Actionable Metabolic Risk By using AIoT, we can turn simple daily observations into smart, actionable insights. We use IoT to pick up lifestyle data and AI to look at it through our models, turning messy data into one clear risk score. This changes the whole focus from just treating a disease once you're sick to actually managing your health while you're well. Instead of waiting for a diagnosis, the user gets a score they can actually use to make changes today. It makes early intervention something that is possible for everyone.

Proposed Solution

Proposed Product

Our solution is a health system that works through two main parts: a physical health portal you can keep at your desk or at home, and a mobile app. It works like a personal early warning tool. We use non-invasive sensors to pick up your body's data along with your daily habits to create what we call a Metabolic Risk Score. The whole goal is to help people, especially those of us balancing busy work and study schedules, to actually understand our own health data so we can take action before problems get serious.

Key Features and Industry 5.0 Principles The heart of the system is our triple-pipeline analysis. Instead of just giving one vague result, the system splits your data into three specialized models. This lets you see your specific risk for fatty liver (NAFLD), type 2 diabetes, and hypertension all at once. It's a way to see the specific details while still understanding your overall health.

Following Industry 5.0 principles, we designed this to be a health partner rather than just a screen that shows numbers. It's built to be human-centric, meaning it gives simple tips you can actually use to make better lifestyle choices or know when it's time to see a doctor. We want to empower people to take charge of their health, not try to replace professional medical advice.

We also focus on making this an integrated hub. By bringing together things like BMI, heart rate, and sleep quality into one dashboard, we break down those old data silos we talked about. This turns messy physiological signals into a clear risk score, shifting health management from just treating an illness once it happens to proactively preventing it.

Uniqueness of the Solution

Connecting the Dots instead of Data Silos Most health trackers people use today only focus on one thing at a time, like your heart rate or how many steps you walked. This leaves the data stuck in separate pieces. Even tools like the PREVENT Calculator or Helsa usually just look at clinical lab results or general wellness. Our solution is different because it looks at how metabolic syndrome is all connected. We are building the only platform that links the risks of your liver, heart, and blood sugar together. Instead of scattered, unrelated numbers, we provide a full picture of your metabolic health.

Predictive Intelligence vs. Just Showing Data A lot of existing apps, like U by Prodia, are just descriptive. They tell you what is happening to your body right now. Our system is predictive. By using AI trained on real clinical data, we can spot high-risk trends before you even feel any symptoms. This directly solves the problem of late diagnosis that we see so often. We move the user from just looking at data to actually understanding their future health risks.

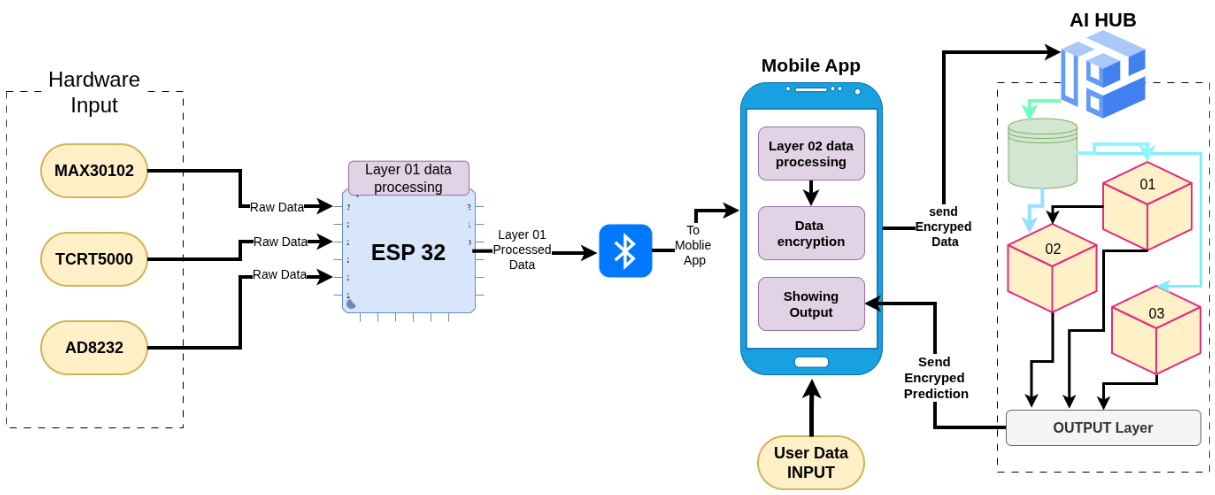
Simple Enough for Everyday Use Many medical tools and advanced tech like NuraLogix can be really intimidating and hard for a regular person to understand. We've simplified everything into a single, color-coded Metabolic Risk Score: Low, Medium, or High. It acts like a simple health mirror at the touch of a button. This makes professional-level health insights easy to understand for busy people who don't have time to study complex medical reports.

Technical Overview and Implementation

Technical Details

The proposed system follows a modular implementation approach that connects physical health monitoring with digital risk analysis. The architecture is organized into three main stages: data collection, intelligent processing, and user feedback. By keeping the system non-invasive and easy to use, it is designed to fit naturally into the daily routine of students and working professionals without disrupting their lifestyle.

Methodology and Data Flow



1. Sensing and User Input

Data collection begins at the Physical Health Portal, where non-invasive sensors are used to capture key physiological signals such as heart activity and blood volume changes. In parallel, users enter basic lifestyle and demographic information—such as age, body measurements, and daily habits—through the mobile application. This combination allows the system to consider both physiological signals and lifestyle factors.

2. Signal Processing and Feature Extraction

The collected raw signals are then processed to remove noise and improve accuracy. This step ensures that sensor data is reliable before analysis. Through this process, raw electronic signals are converted into meaningful features, such as heart rate trends and vascular indicators, which can be used for risk assessment.

3. Parallel Risk Assessment

At the core of the system is a triple-pipeline analysis framework. Instead of applying a single general model, the refined data is analyzed simultaneously using three separate machine learning models. Each model is designed to estimate early risk indicators for Non-Alcoholic Fatty Liver Disease (NAFLD), Type 2 Diabetes, and Hypertension. Evaluating these conditions together allows the system to capture their interrelated nature and identify patterns that may be missed by isolated assessments.

4. Insight Delivery and User Feedback

The final outputs are summarized into clear risk categories and presented to the user through a unified dashboard. A simplified Metabolic Risk Score is displayed along with practical, easy-to-understand guidance aimed at improving overall wellness. This approach helps users move from simply viewing data to understanding and acting on their health information.

User Scenario

Scenario: Early Risk Awareness During a Busy Workday

Saman is a 40-year-old working professional in Colombo who rarely finds time for routine medical check-ups. Although he feels generally healthy, his sedentary desk job, high work stress, and irregular sleep patterns place him at increased metabolic risk.

He begins using the proposed Health Portal at his workplace, placing his finger on the non-invasive sensors for approximately 30 seconds each morning. While he continues his daily routine, the system captures cardiovascular signals, while the mobile application tracks sleep patterns and BMI trends over time.

After two weeks of sustained work stress and poor sleep, the system detects rising cardiovascular load and unfavorable metabolic patterns. Even in the absence of noticeable symptoms, the application flags a moderate risk level for hypertension and type 2 diabetes. Rather than waiting for a clinical diagnosis, Saman follows the app’s personalized lifestyle guidance to improve his physical activity and sleep habits.

By identifying these early, “silent” risk indicators, the system supports timely lifestyle adjustments that help reduce long-term metabolic risk before it progresses into chronic disease.